3. Review: modeling two species populations

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Modeling two populations

Start of transcript. Skip to the end.

We will now be interested in modelling how a second

population, y, interacts with our population x. Remember

that x represented a population of deer. For example, y

could represent a population of moose. Deer and moose both

eat the same food, i.e. leaves and vegetation, and so x and

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The basic model for two interacting populations is the 2×2 autonomous system of **Lotka-Volterra equations** :

$$\dot{x} = (ax - bx^2) \pm cxy \tag{5.3}$$

$$\dot{y} = (dy - hy^2) \pm fxy. \tag{5.4}$$

where the equations govern the rates of change of the populations x and y respectively, and a, b, c, d, f, h are the 6 parameters of the system. Note that here we have written the equations so that the parameters are positive.

In each equation, the two terms in the brackets model the **logistic** growth of a single population in the absence of the other.

Interactions between the two populations

The final term in each equation is proportional to the product $xy,$ and models the effect of interaction between x and y on the respective population. For example, in the first equation, which governs the rate of change of the population $x,$ if the term proportional to xy is positive, then $\dot{x}$ is increased. This means interactions are beneficial to the x population.

The signs of the xy terms in the two equations determine which of the following scenarios are modeled.

1. **Competition**:

$$\dot{x} = (ax - bx^2) - cxy \tag{5.5}$$

$$\dot{y} = (dy - hy^2) - fxy. \tag{5.6}$$

   If the terms proportional to xy in both equations are negative, then interactions decrease the rates of change of both x and y. This models two species mutually harmful to one another. An example is moose and deer competing for vegetation in the same habitat.

2. **Mutualism**:

$$\dot{x} = (ax - bx^2) + cxy \tag{5.7}$$

$$\dot{y} = (dy - hy^2) + fxy. \tag{5.8}$$

   If the xy terms in both equations are positive, then interactions increase the rates of change of both x and y. This models two species mutually beneficial to one another. An example is human and the bacteria living in our digestive system.

3. **Predator-prey**:

Calculator

$$\dot{x} = \left(ax - bx^2\right) - cxy \tag{5.9}$$

$$\dot{y} = \left(dy - hy^2\right) + fxy. \tag{5.10}$$

If the signs of the xy terms in the two equations are different, then interaction benefits one population but harms the other. This models a predator species and a prey species. An example is wolves and deer, with the deer being a food source for the wolves.

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